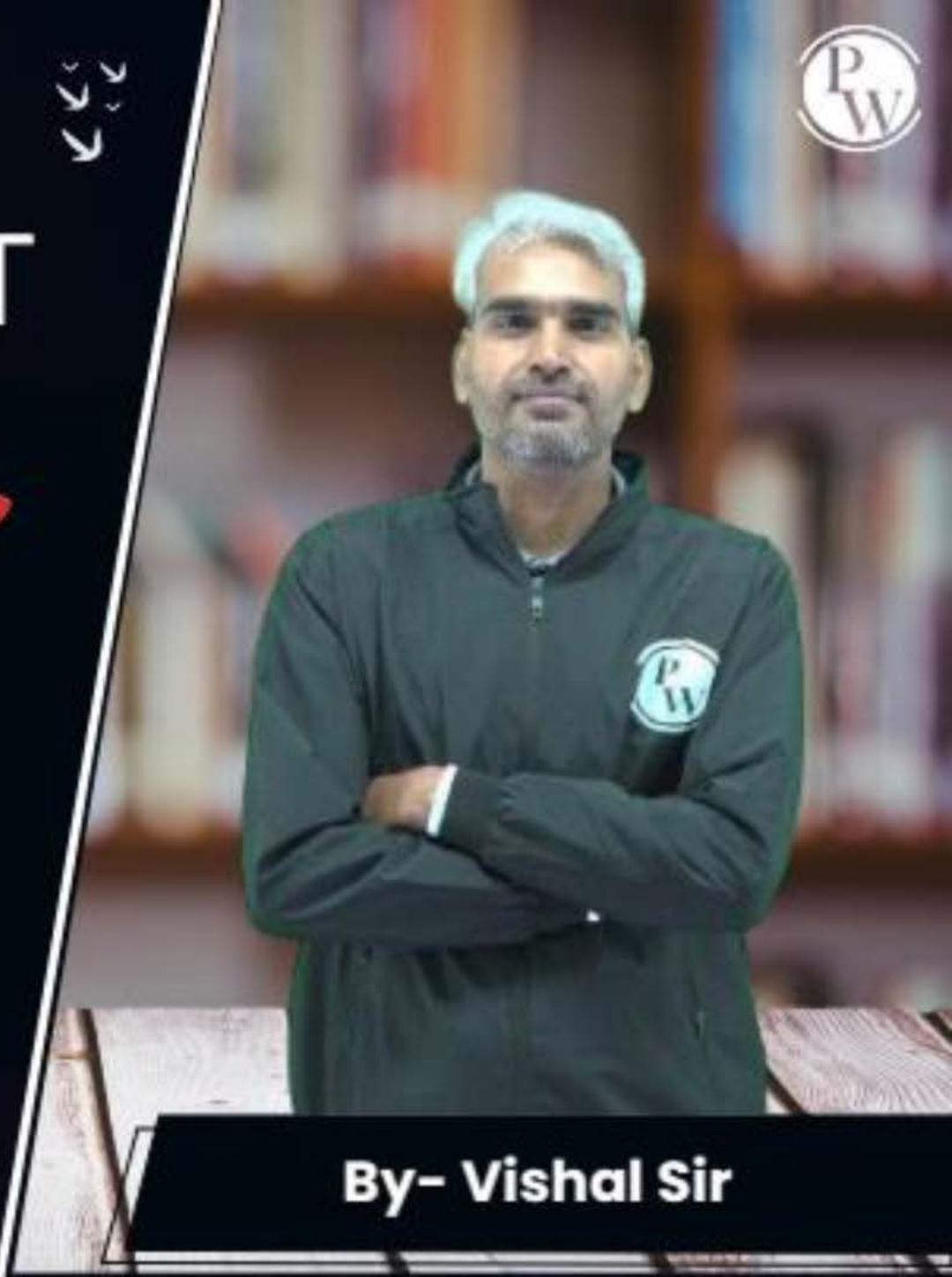
Computer Science & IT



Discrete Mathematics

Graph Theory

Lecture No. 09



By- Vishal Sir

Recap of Previous Lecture



- Topic: Vertex coloring
- Topic: Chromatic number
- Topic: Edge coloring
- Topic: Matching

Slide

Topics to be Covered



Topic Line covering

Topic Line independent set

Topic Vertex covering

Topic Vertex independent set

Next Class

Slide



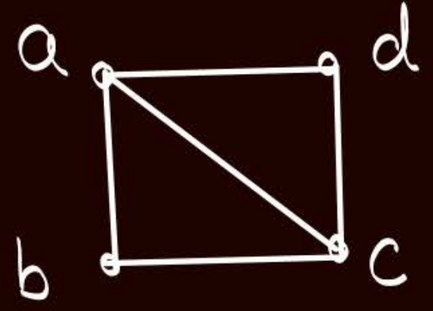
Topic : Matching

A subgraph M of graph G is called a matching of graph G, if all the vertices of graph G are incident with at most one edge in subgraph M,

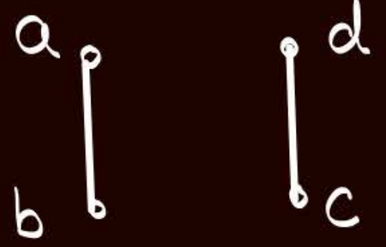
↳ All vertices of graph G must be present in sub-graph M

In a Matching of graph G, $\deg(v) \leq 1 \quad \forall v \in G$

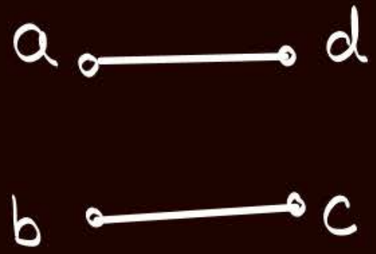
eg:



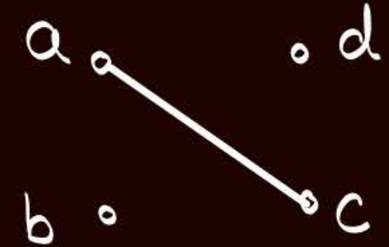
G1



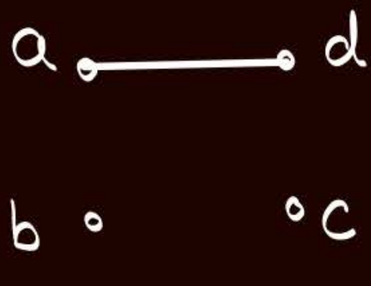
"M_1"



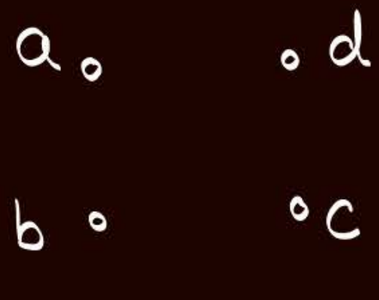
"M_2"



"M_3"



"M_4"



"M_5"

In a matching of graph G1, adjacent edges are not allowed.

M_1, M_2, M_3, M_4, M_5 all are matching of graph G1

Other matchings are also possible for graph G1.

Note:

① In a Matching of graph G.
if $\deg(v) = 1$, then vertex 'v' is called a Matching Vertex

② In a Matching of graph G.
if $\deg(v) = 0$, then vertex 'v' is called a non-Matching Vertex

Maximal

↓

Nothing can be added, without destroying the Property

Minimal

↓

Nothing can be removed, without destroying the Property



Topic : Maximal Matching

- A matching of graph G is called a maximal matching, if no other edge of graph G can be added to that matching without destroying its property of being a matching.

In above example,

M_1, M_2 and M_3 are maximal matching of graph G.



Topic : Maximum Matching / Largest Maximal Matching



- The matchings of graph G with maximum number of edges are called maximum matching of graph G.

{ There may be more than one maximum matching for the given graph }

In the above example,

M_1 & M_2 are maximum matching

{ M_3 is maximal matching but not a maximum matching }

All maximum matching of graph G are also maximal matching of graph G, but all maximal matchings need not be maximum matching



Topic : Matching Number

* Matching Number of Graph G = Number of edges in any one of the maximum matching of graph G.

In the above example,

Matching no. of G = 2

↑ w.r.t. M_1 or M_2

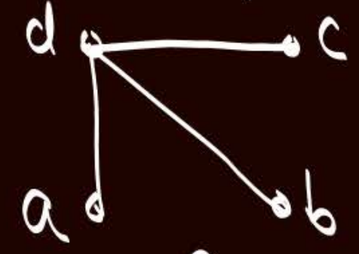
Note:-

Let G be a graph with n-vertices, then

matching number of graph G is $\leq \left\lfloor \frac{n}{2} \right\rfloor$

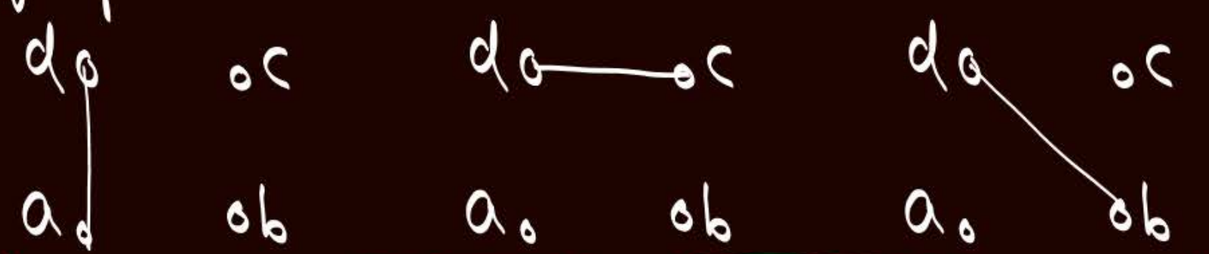
↑ it is the upper bound on matching No.

eg: Consider the following graph



G

in graph G, $n=4$



All are maximum matching of graph G

∴ Matching No. of $G = 1 < \left\lfloor \frac{4}{2} \right\rfloor$

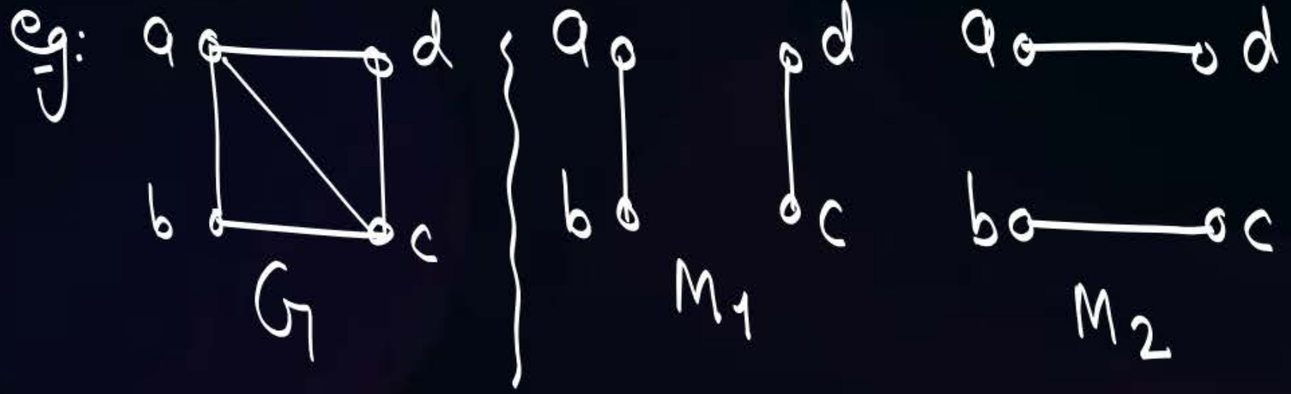


Topic : Perfect Matching

* A matching of graph G is called a perfect matching if and only if degree of each vertex in the matching of graph G is '1'.

in a perfect matching, $deg(v) = 1, \forall v \in G$

eg:



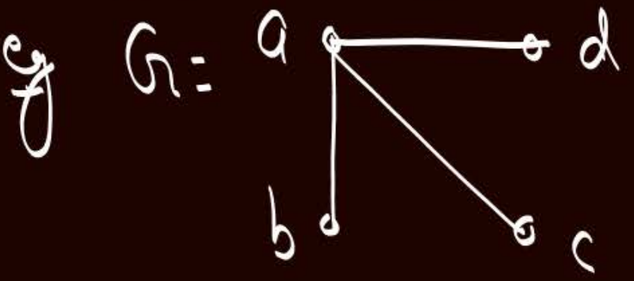
Both M_1 & M_2 are Perfect matching of graph G.

Note :-

① Every perfect matching of graph G, is also a maximum matching of graph G.

② But, Every maximum matching of graph G need not be a perfect matching

eg G :



} Maximum matching exist for graph G, but no perfect matching

Note:-

If perfect matching exists for graph G, then number of vertices in graph G are even.

(Converse of the statement need not be true)

Note:-

Let G be a graph with n-vertices,

If perfect matching exists for graph G, then n = even.

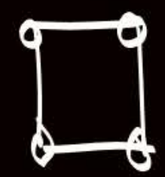
and Matching Number of graph $G = \frac{n}{2}$

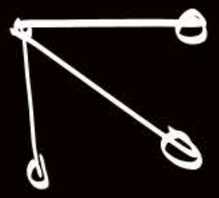
Note:-

① Perfect matching need not exist for Every graph

② In a graph G,
If no. of vertices are odd, then perfect matching
can never exist for that graph

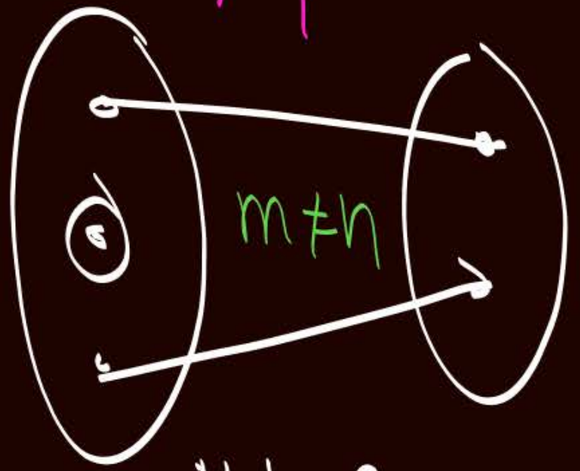
③ In a graph G,
If no. of vertices are even, then Perfect matching
may or may not exist for that graph

eg. 

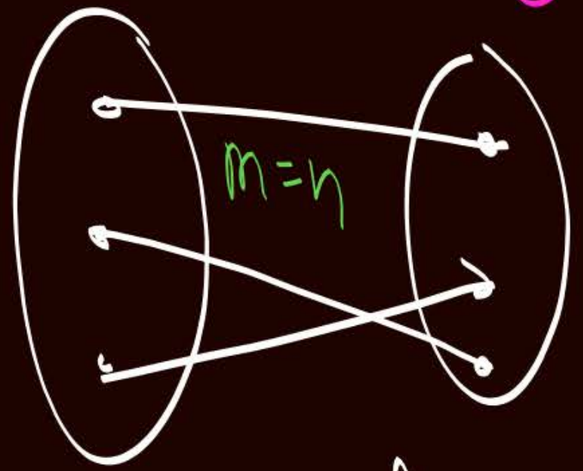
eg. 

- For a Complete graph K_n Perfect matching exists if and only if n = Even in K_n

- In a Complete bipartite graph $K_{m,n}$ Perfect matching exist if and only if $m = n$

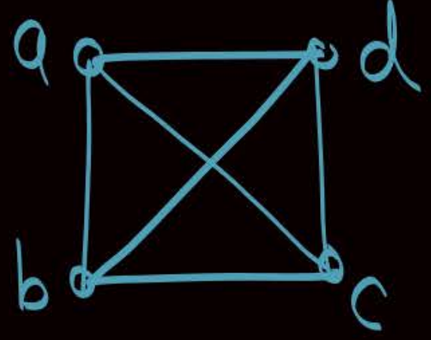


Not a Perfect matching

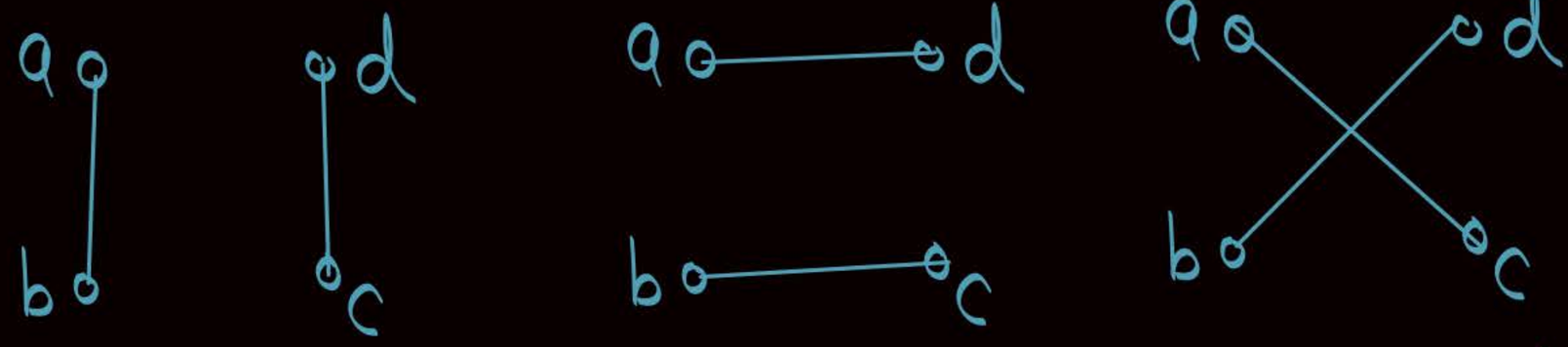


it is Perfect matching

* Let K_n is a Complete graph, where 'n' is Even.
Find the **no. of Perfect matching** in Complete graph K_n

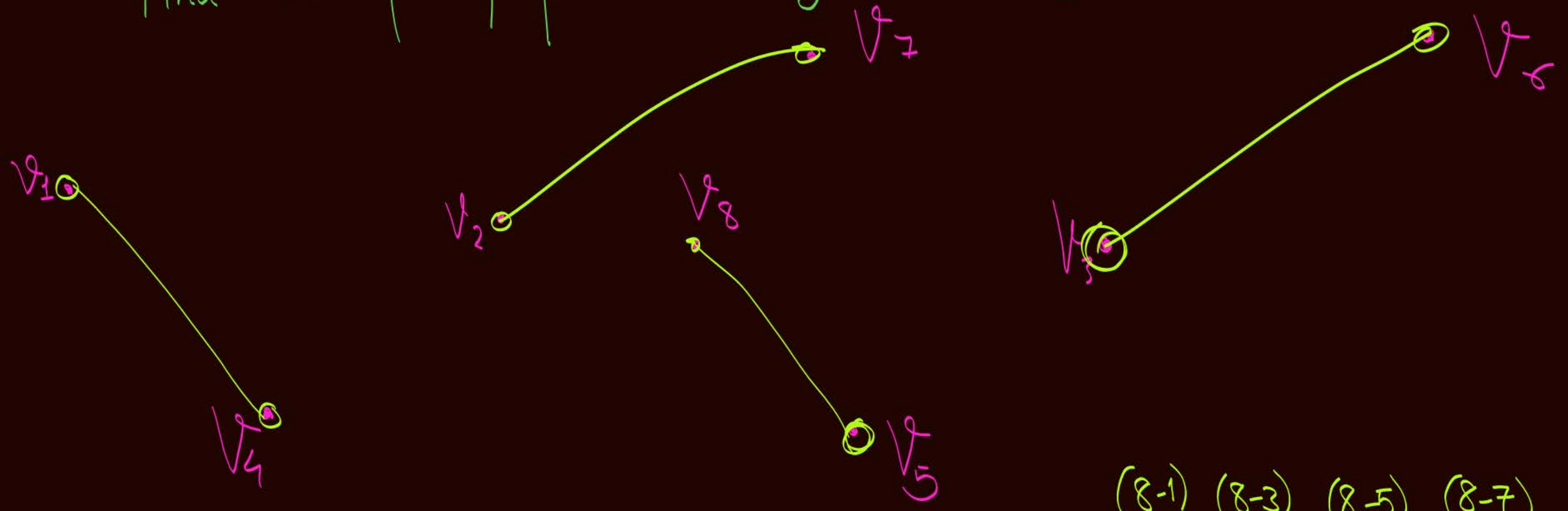


K_4



These are the only three Perfect Matchings Possible for K_4

Find no of perfect matchings in K_8



(8-1) (8-3) (8-5) (8-7)

no of perfect matchings in $K_8 = 7 * 5 * 3 * 1$

$= \underline{105}$

Number of Perfect matching in K_n {n = Even}

Number of Perfect matching in K_n

$$= (n-1)(n-3)(n-5)\ldots\ldots 5\cdot 3\cdot 1$$

$$= (n-1)(n-3)(n-5)\ldots\ldots 5\cdot 3\cdot 1 * \frac{n(n-2)(n-4)\ldots 6\cdot 4\cdot 2}{n(n-2)(n-4)\ldots 6\cdot 4\cdot 2}$$

$$= \frac{n(n-1)(n-2)(n-3)\ldots\ldots 6\cdot 5\cdot 4\cdot 3\cdot 2\cdot 1}{n\ (n-2)(n-4)\ldots 6\cdot 4\cdot 2}$$

$$= \frac{n!}{(2\cdot\frac{n}{2})\cdot(2\cdot(\frac{n}{2}-1))(2(\frac{n}{2}-2))\cdot(2\cdot 3)\cdot(2\cdot 2)\cdot(2\cdot 1)}$$

$$= \frac{n!}{(2)^{n/2}\cdot\{(\frac{n}{2})\cdot(\frac{n}{2}-1)\cdot(\frac{n}{2}-2)\ldots 3\cdot 2\cdot 1\}} = \boxed{\frac{n!}{(2)^{n/2}\cdot(\frac{n}{2})!}}$$

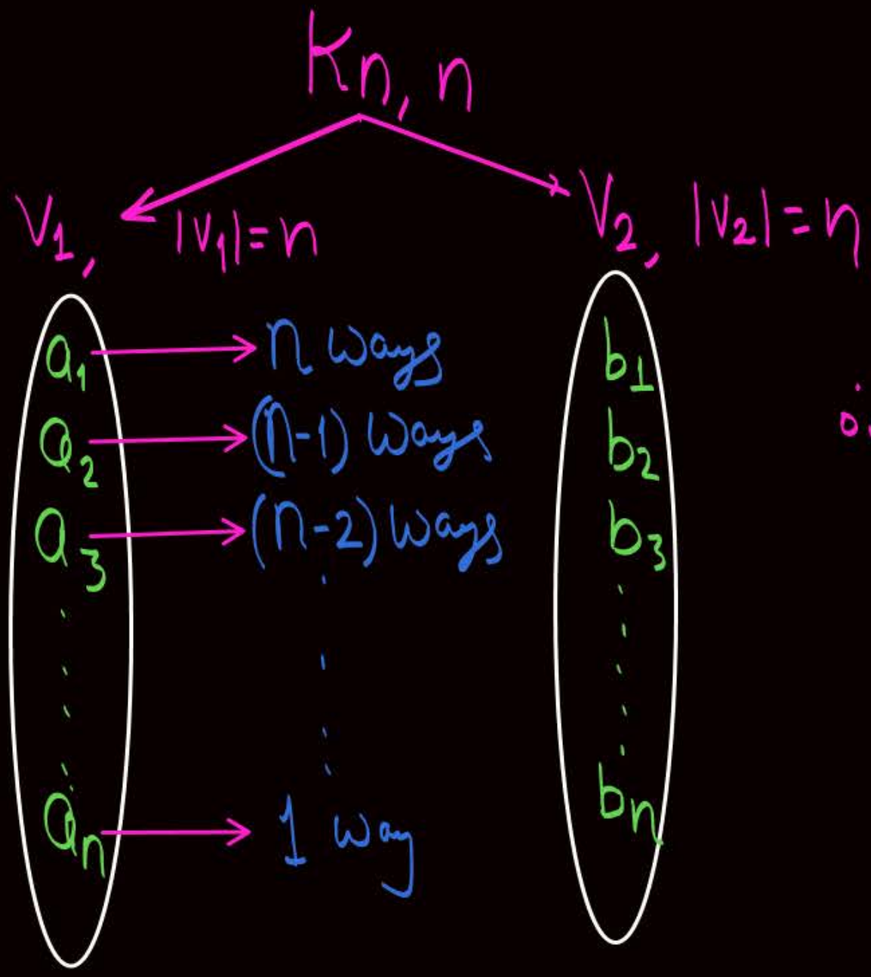
∴ Number of Perfect matching in K_{2n}

it will always be even

just replace 'n' by '2n'

$$\therefore \text{Number of Perfect matching in } K_{2n} = \frac{(2n)!}{(2)^{\frac{2n}{2}} \cdot \left(\frac{2n}{2}\right)!} = \boxed{\frac{(2n)!}{2^n \cdot n!}}$$

Q:- Find the number of perfect matching in Complete bi-partite graph $K_{n,n}$



∴ # Perfect matchings in Complete bipartite graph $K_{n,n}$ $= n(n-1)(n-2)\ldots 3 \cdot 2 \cdot 1$

$= n!$

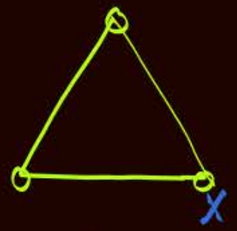
Today's Topic

H.W

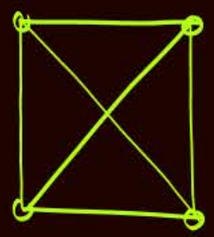
Q:- Find the matching number of Complete graph K_n



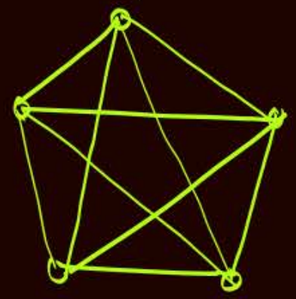
K_2



K_3



K_4



K_5

Matching No. of $K_2 = 1$

Matching No. of $K_3 = 1$

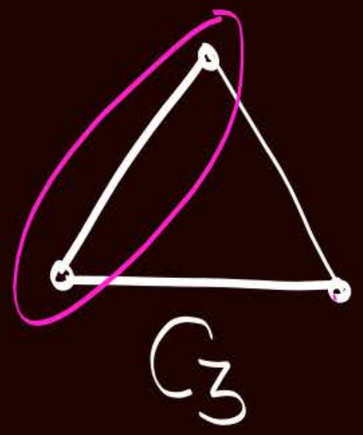
Matching No. of $K_4 = 2$

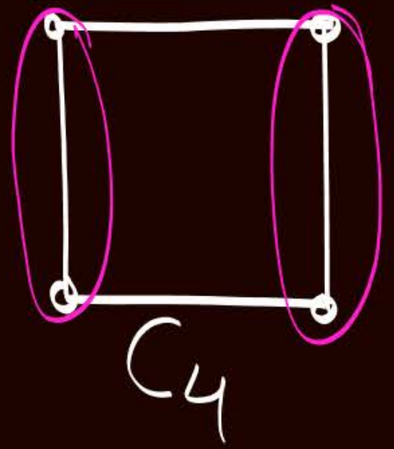
Matching No. of $K_5 = 2$

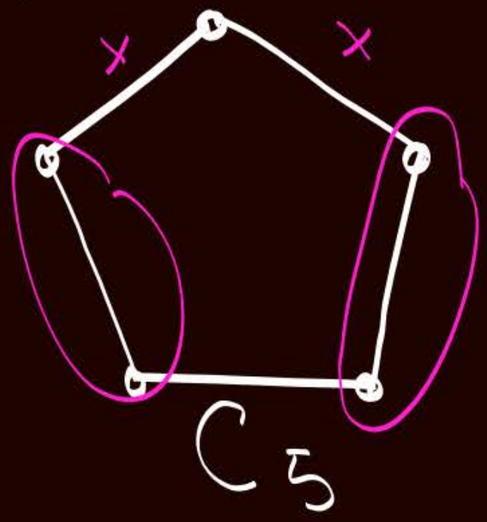
Matching no. of $K_n = \left\lfloor \frac{n}{2} \right\rfloor$

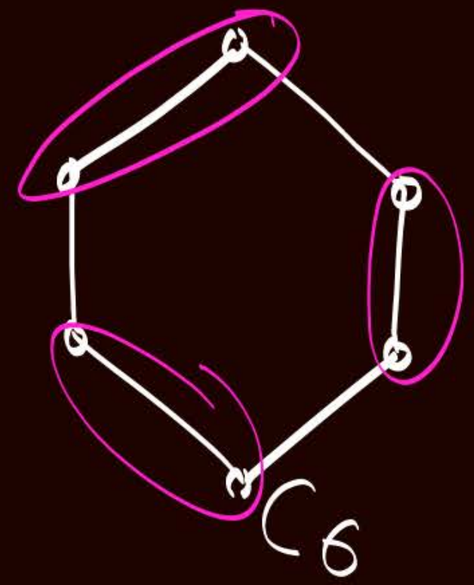
Q:: Find matching number of Cycle graph C_n

H.W.









Matching No. = 1

Matching No. = 2

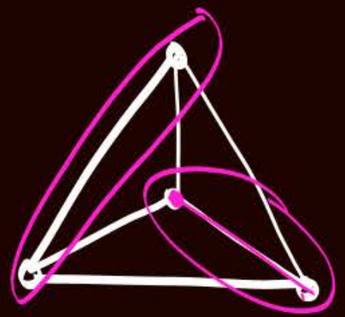
Matching No. = 2

Matching No. = 3

Matching no. of C_n : $\left[\frac{n}{2}\right]$

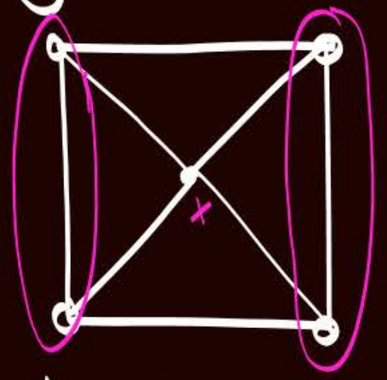
Q:. Find matching number of Wheel graph W_n

H.W.



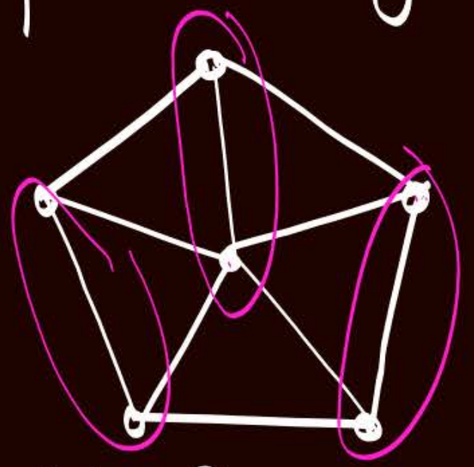
$W_4 = C_3 + 1$

Matching No. = 2



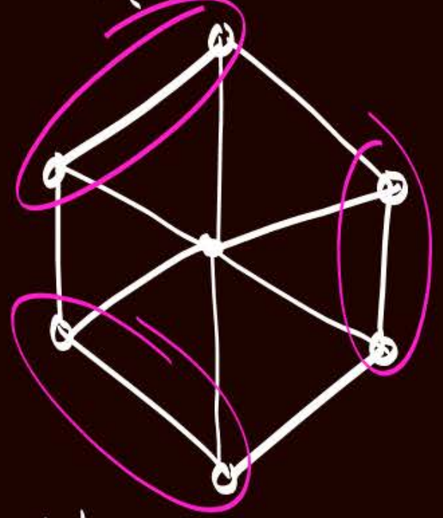
$W_5 = C_4 + 1$

Matching No. = 2



$W_6 = C_5 + 1$

Matching No. = 3

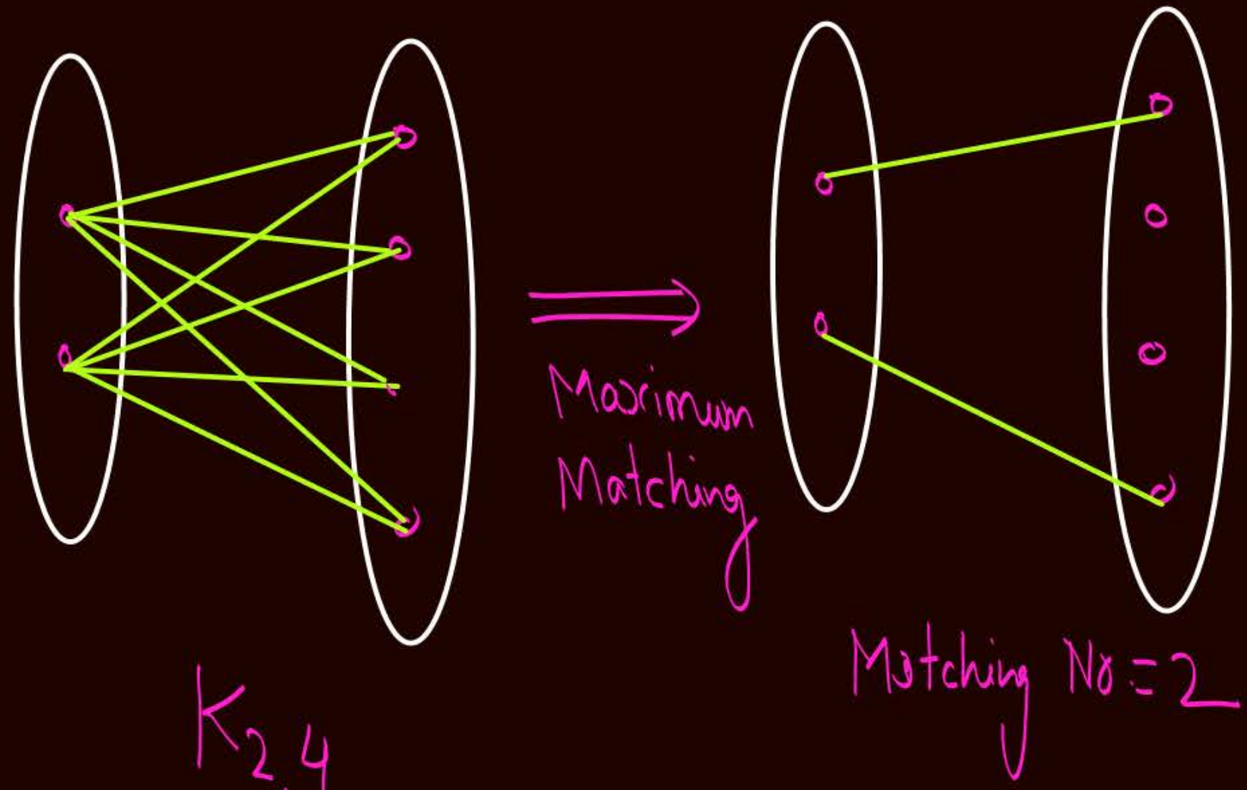


$W_7 = C_6 + 1$

Matching No. = 3

Matching no. of $W_n = \lfloor \frac{n}{2} \rfloor$

Q:- Find the matching number of Complete bipartite graph $K_{m,n}$



Matching number of Complete bipartite graph $K_{m,n} = \text{Min}(m,n)$

Q:- Find the matching number of Star graph with 'n' vertices.

t.w.i.

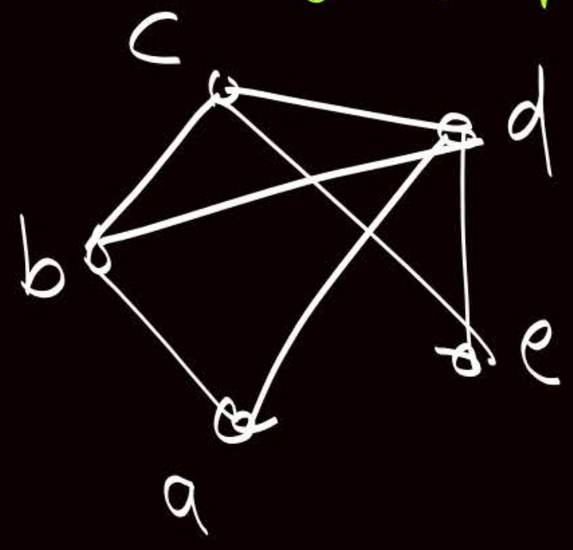
Star graph with n-vertices $\equiv$ $K_{1,n-1}$ or $K_{n-1,1}$

$\therefore$ Matching No. $= \min(1, n-1)$

$= 1$

* Note:- Adjacency matrix can be used to represent any graph.

eg.



G

$\Rightarrow$ Adjacency matrix of G =

	a	b	c	d	e
a	0	1	0	1	0
b	1	0	1	1	0
c	0	1	0	1	1
d	1	1	1	0	1
e	0	0	1	1	0

Note:- In the adjacency matrix of Every Perfect matching there will be exactly one '1' in each row and there will be exactly one '1' in each column, and in that case inverse of adjacency matrix will be adjacency matrix itself.

Covering

Covering

Line Covering / Edge Covering

We will try to Cover all the vertices of the graph using subset of edges

Vertex Covering

We will try to Cover all the edges of the graph using subset of vertices



Topic : Line/Edge Covering

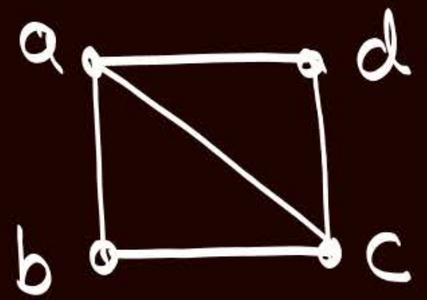
Let G=(V,E) be a graph.

A subset C of set of edges E is called line covering of graph G if every vertex of the graph is incident with at least one edge of subset C.

i.e, in a sub-graph corresponding to the line covering

$$\deg(v) \geq 1, \quad \forall v \in G$$

eg:-

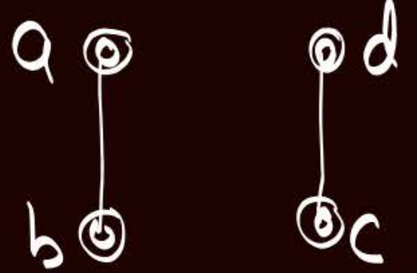


"G"

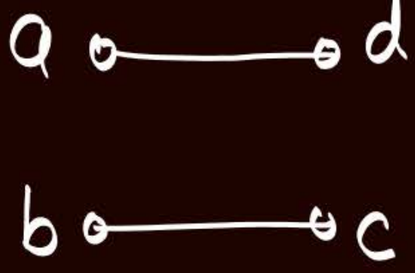
$V = \{a, b, c, d\}$

$E = \{\{a,b\}, \{a,c\}, \{a,d\}, \{b,c\}, \{c,d\}\}$

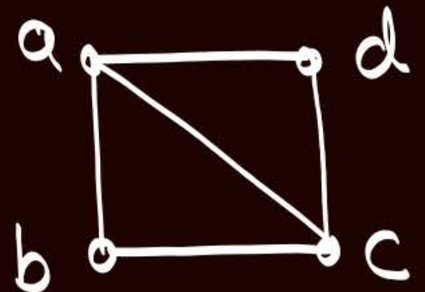
$C_1 = \{\{a,b\}, \{c,d\}\}$



$C_2 = \{\{a,d\}, \{b,c\}\}$



eg:-

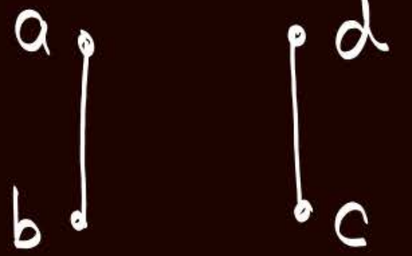


"G"

$V = \{a, b, c, d\}$

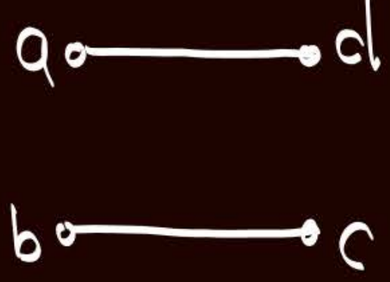
$E = \{\{a,b\}, \{a,c\}, \{a,d\}, \{b,c\}, \{b,d\}\}$

$C_1 = \{\{a,b\}, \{c,d\}\}$

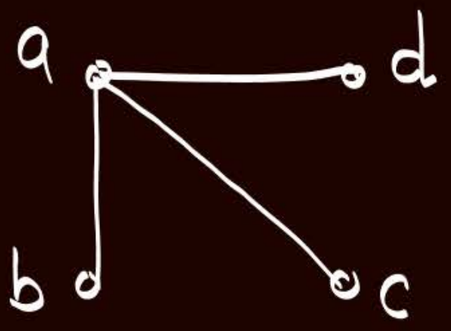


"C_1"

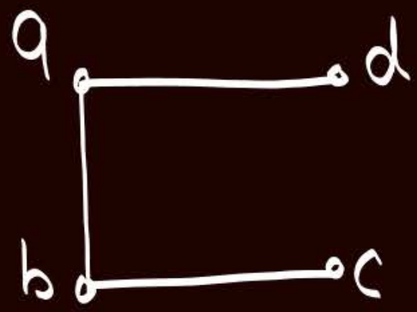
$C_2 = \{\{a,d\}, \{b,c\}\}$

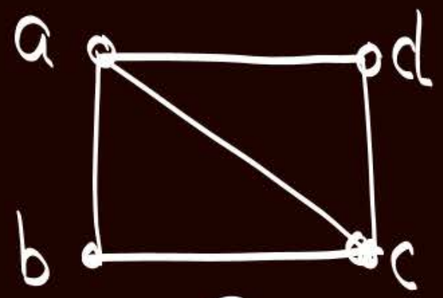


"C_2"



$C_3 = \{\{a,b\}, \{a,c\}, \{a,d\}\}$

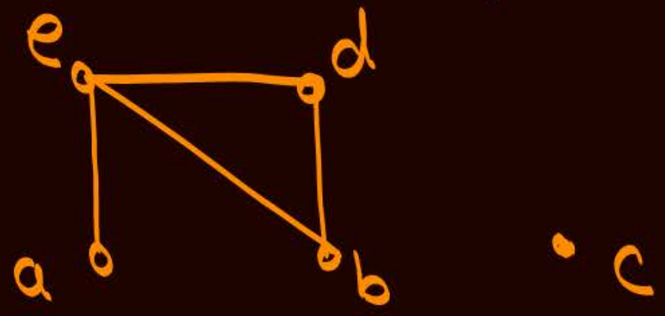




$C_4 = \{\{a,b\}, \{b,c\}, \{a,d\}\}$ $\quad C_5 = E$

C_1, C_2, C_3, C_4, C_5 etc.. are line Covering of graph G

* Consider the following graph G.

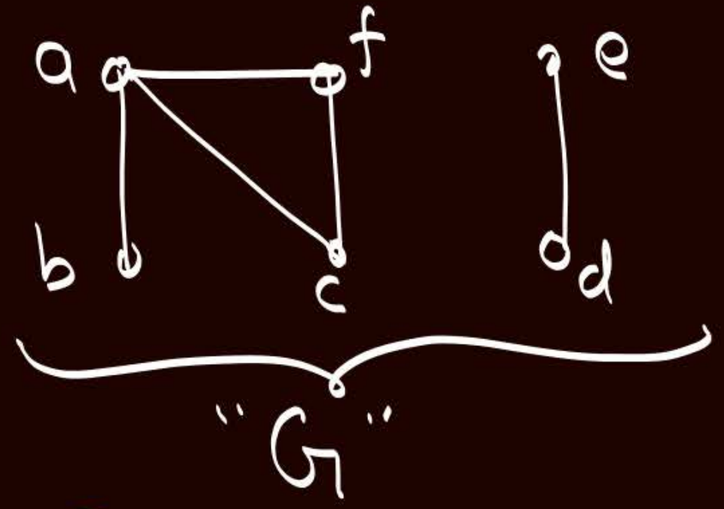


do we have any line covering for graph G.

Ans: . No line covering exist for the given graph

> Line covering does not exist for a graph G if and only if any isolated vertex exist in graph G

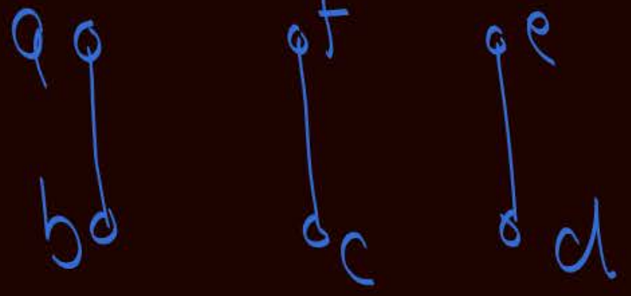
eg:



← disconnected graph

Does line covering exist for graph G.

Ans = Yes.



{ Line covering can exist for a disconnected graph as well, Provided no isolated vertex exist.



Topic : Minimal line covering

A line covering of graph G is said to be minimal if no edge can be deleted from that line covering without destroying its property to cover all the vertices of the graph.

In the above example,

C_1, C_2 & C_3 are minimal line covering



Topic : Minimum line covering

Smallest minimal line Covering

- Line coverings of graph G with minimum number of edges are called minimum line coverings of graph G.

In the above example,

C_1 & C_2 are minimum line covering

C_3 is minimal, but not minimum

Note:- ① Every minimum line covering of graph G is also a minimal line covering, but every minimal line covering need not be a minimum line covering {eg. C_3}

② Minimal line covering of graph G can never contain any cycle

③ In a minimal line covering of graph G there can not be any path of length '3' or more



Topic : Line covering number (α_1)

Line covering no. of graph G = No. of edges in any one of the minimum line covering

α_1 = No. of edges in any one of the minimum line covering

In the above example,
Line covering No of $G = 2$

For any graph G with n-vertices
line covering No. of $G \geq \left\lceil \frac{n}{2} \right\rceil$



Topic : Line independent set

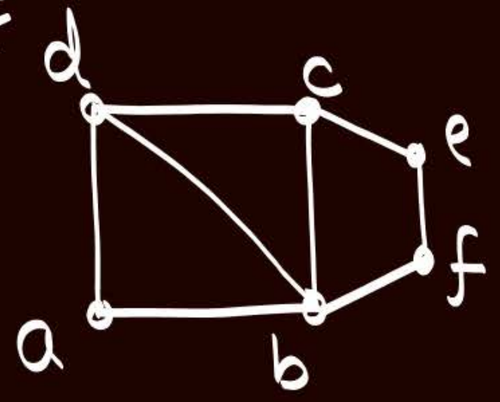
- Let $G = (V, E)$ be a graph.

A subset 'L' of set of edges 'E' is called a line independent set if no two edges of subset L are adjacent to each other.

Because in a line independent set no two edges can be adjacent
∴ In a subgraph corresponding to line independent set, $\deg(v) \leq 1, \forall v \in G$

We know $\deg(v) \leq 1, \forall v \in G$ is also the condition for matching
Hence Every line independent set of graph G is also a matching of graph G.

eg:



G

$L_1 = \{\{a,b\}\}$

A subset of edges with a single edge is always a line independent set.

$L_2 = \{\{a,b\}, \{c,d\}\}$

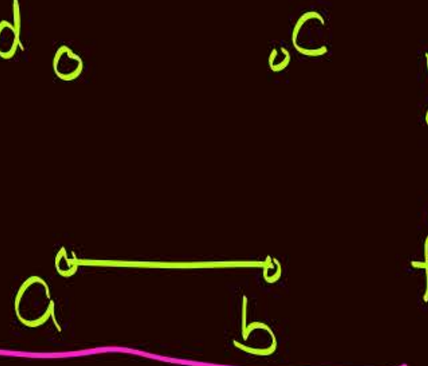
$L_3 = \{\{a,d\}, \{e,f\}\}$

$L_4 = \{\{b,d\}, \{c,e\}\}$

$L_5 = \{\{a,d\}, \{b,c\}, \{e,f\}\}$

$L_6 = \{\{a,b\}, \{c,d\}, \{e,f\}\}$

Matching w.r.t L_1 =



All subsets of edges from L_1 to L_6 are Line independent sets of graph G.



Topic : Maximal Line independent set

- A Line independent set is called a maximal line independent set if no new edge from the graph can be added to the set without destroying its property of being independent set.

In the above example,

L_4, L_5 & L_6 are maximal line independent set



Topic : Maximum Line independent set

Largest maximal line independent set

* Line independent sets with maximum number of edges are called maximum line independent sets.

In the above example,

L_5 & L_6 are maximum line independent set

* Every maximum line independent set is also a maximal line independent set, but Every maximal line independent set need not be a maximum line independent set. { e.g. L_4 }



Topic : Line independence number (β_1)

Line independence Number of graph G = Number of edges in any one of the maximum line independent set of G.

β_1 = No. of edges in any one of the maximum line independent set = Matching No. of graph G

For a graph G with 'n' vertices
Line independence no. of graph G $\leq \left\lfloor \frac{n}{2} \right\rfloor$

In the above example,

Line independence No. of graph $G_7 = 3$

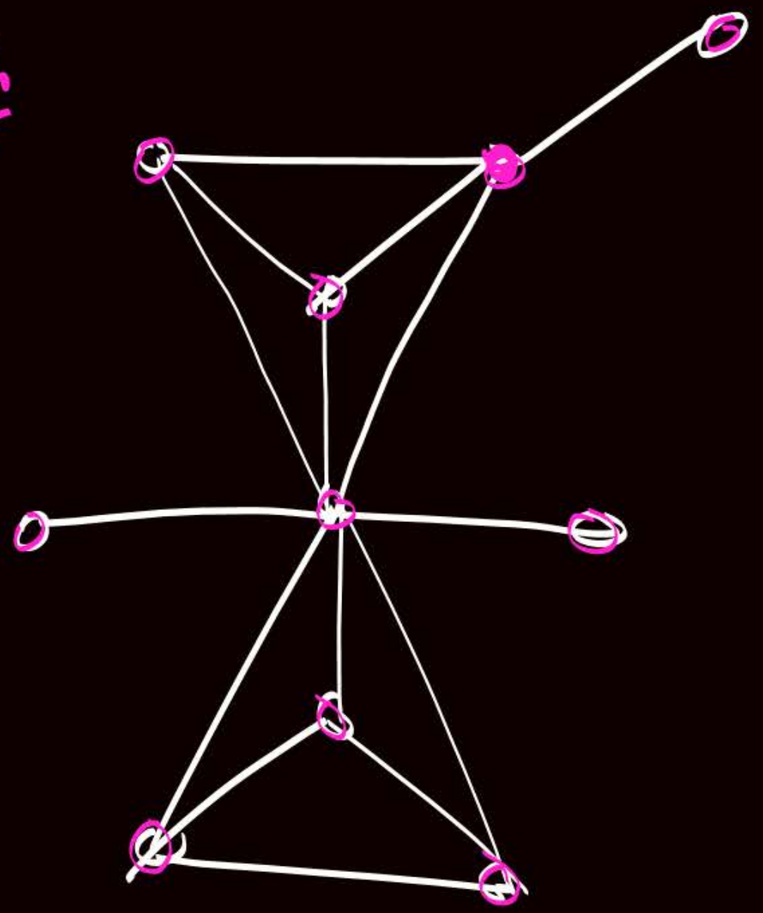
w.r.t L_5 or L_6

* For any graph $G_1 = (V, E)$

$$\alpha_1 + \beta_1 = |V(G_1)|$$

i.e.) (Line covering number of graph G_1 + Line independence number of graph G_1) = Total number of vertices in graph G_1

Q:

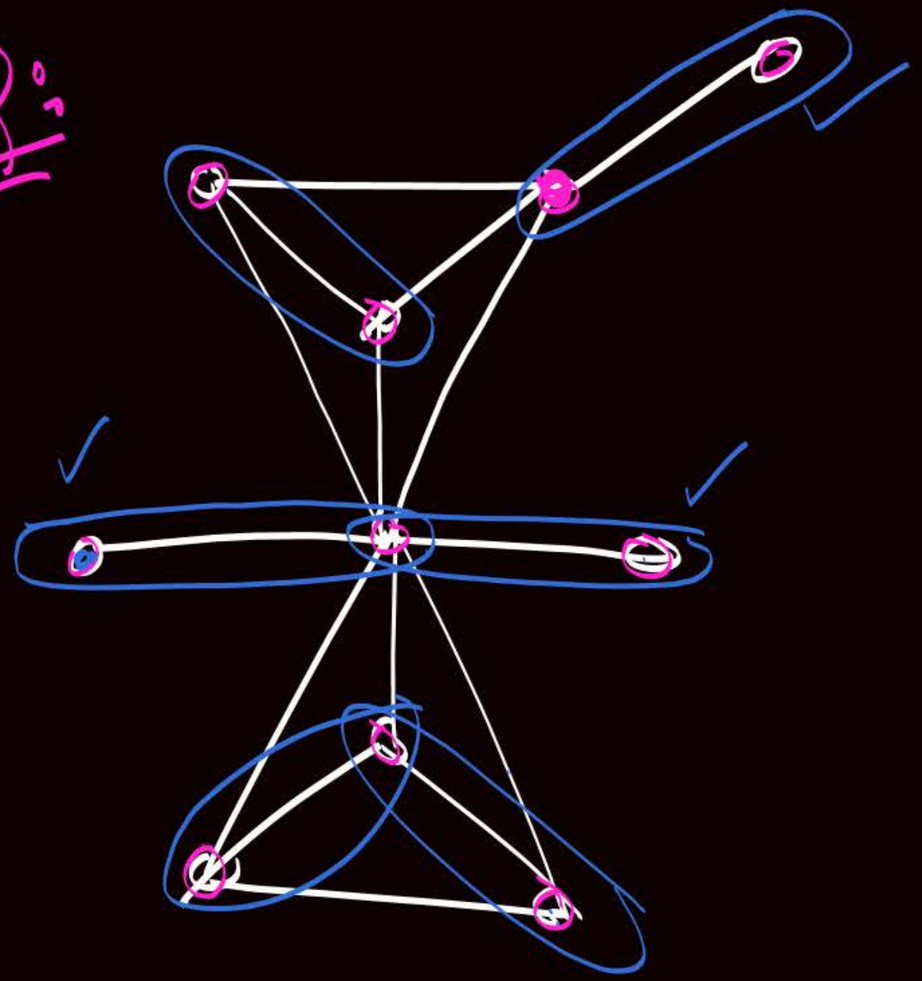


Find α_1 & β_1 for given graph

$\alpha_1 =$

$\beta_1 =$

Q:

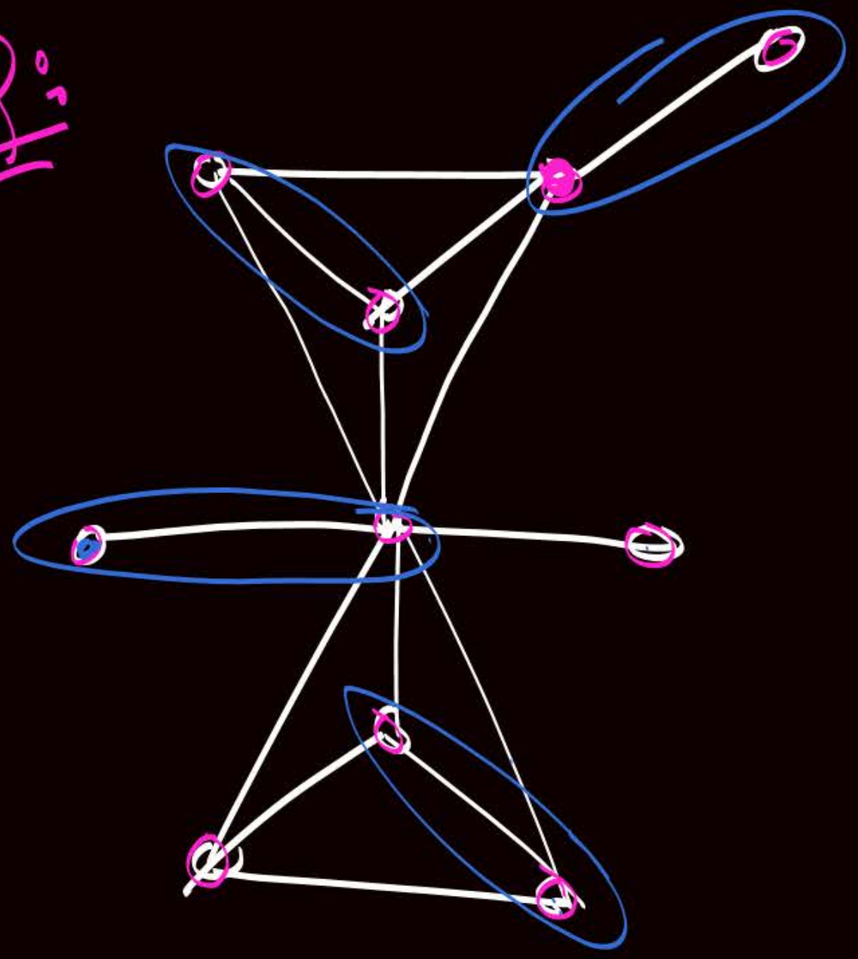


Find α_1 & β_1 for given graph

$\alpha_1 = 6$

$\beta_1 =$

Q:



Find α_1 & β_1 for given graph

$\alpha_1 = 6$

$\beta_1 = 4$

H.W.

Find α_1 & β_1 for the following graphs

(i) Complete graph K_n

(ii) Cycle graph C_n

(iii) Wheel graph W_n

(iv) Complete bipartite graph $K_{m,n}$

(V) Star graph with 'n' vertices

α_1 = line covering no.

β_1 = line independence no.

	Line Covering No. α_1	Line independence No. β_1 =
① Complet graph (K_n)	$\lceil \frac{n}{2} \rceil$	$\lfloor \frac{n}{2} \rfloor$
② Cycle graph (C_n)	$\lceil \frac{n}{2} \rceil$	$\lfloor \frac{n}{2} \rfloor$
③ Wheel graph (W_N)	$\lceil \frac{n}{2} \rceil$	$\lfloor \frac{n}{2} \rfloor$
④ Complete Bipartite graph ($K_{m,n}$)	$\text{Max}(m,n)$	$\text{Min}(m,n)$
⑤ Star graph with n-vertices	$(n-1)$	1

H.W :→ ① Vertex Covering

② Vertex independent set

③ Relationship b/w Vertex Covering
Vertex independent set.

④ Connectivity.



2 mins Summary

Topic Line covering

Topic Line independent set

Topic Vertex covering

Topic Vertex independent set

Slide

THANK - YOU